ABESIT
Group Of Institutions

Project Id:(2023CSEPID47)

Project Synopsis

on

DecentraPay: A Secure Crypto Transaction System

Submitted in Partial Fulfillment of the Requirement

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

By

Gauri Sharma (2302900100089)

Avni Tyagi (2302900100070)

Akshat Singh Aswal (2302900100026)

Charu Chaudhary (2302900100075)

Under the Supervision

of

Mrs. Madhavi Joshi

Assistant Professor

Department of Computer Science & Engineering

ABES INSTITUTE OF TECHNOLOGY, GHAZIABAD



AFFILIATED TO

Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

(ODD SEM, 2025-26)

Table of Contents

Chapter No.	Contents	Page No.
1.	Abstract	1
2.	Introduction	2
3.	Literature Survey	3 - 5
4.	Problem Statement	6
5.	Proposed Methodology	7 - 9
6.	Conclusion	10
7.	References	11

Abstract

The rapid evolution of blockchain technology and Web 3.0 has transformed the way digital transactions are performed by enabling secure, transparent, and decentralized financial systems. Traditional payment systems rely heavily on centralized authorities such as banks and payment gateways, which often result in higher transaction costs, longer processing times, security concerns, and dependence on intermediaries. These limitations have encouraged the adoption of decentralized applications (DApps) that allow users to interact directly through blockchain networks without relying on third-party institutions.

The proposed project, "DecentraPay: A Blockchain-Based Secure Cryptocurrency Transaction System Using Smart Contracts**,"** aims to develop a decentralized application that enables users to perform secure peer-to-peer cryptocurrency transactions using the Ethereum blockchain. The application integrates MetaMask wallet authentication, allowing users to securely connect their blockchain wallets and initiate transactions directly from the web interface.

The frontend of the application is developed using React.js, providing a responsive and user-friendly interface, while Ethers.js serves as the communication layer between the frontend and the Ethereum blockchain. Transaction processing is handled through smart contracts written in Solidity, which automatically execute predefined conditions without requiring any intermediary. The smart contracts securely validate, record, and execute cryptocurrency transactions while maintaining complete transparency and immutability on the blockchain.

The project is developed and tested using the Hardhat development environment, which facilitates smart contract compilation, deployment, debugging, and testing. Once deployed, every successful transaction is permanently recorded on the blockchain, allowing users to view transaction history with complete transparency and security.

The proposed system demonstrates the practical implementation of blockchain technology by integrating decentralized finance principles with modern web technologies. It provides improved security, reduced transaction costs, faster settlement, enhanced transparency, and greater user control over digital assets. The project serves as an educational implementation of Web 3.0 concepts while showcasing the real-world applications of decentralized applications and smart contracts in modern financial systems.

Introduction

The emergence of blockchain technology has significantly transformed digital transactions by introducing decentralized and trustless systems that eliminate the need for centralized authorities. With the increasing popularity of cryptocurrencies such as Bitcoin and Ethereum, there has been a growing demand for applications that enable users to securely transfer digital assets without depending on banks or financial intermediaries. This shift has led to the development of Web 3.0, the next generation of the internet.

Traditional online payment systems operate under centralized control, where financial institutions validate and process every transaction. Although these systems are widely adopted, they often involve high transaction fees, delayed settlements, limited transparency, and potential security vulnerabilities. Blockchain technology overcomes these limitations by maintaining a distributed ledger across multiple nodes, where every transaction is verified through consensus mechanisms and permanently stored in an immutable ledger. This ensures greater transparency, security, and resistance to fraud while eliminating the dependence on intermediaries.

One of the most significant innovations introduced by blockchain technology is the concept of smart contracts. Smart contracts are self-executing programs deployed on blockchain networks that automatically perform predefined operations when specified conditions are met. These contracts eliminate manual intervention and enable secure, transparent, and reliable execution of digital transactions.

The proposed project, DecentraPay, is a decentralized application (DApp) designed to facilitate secure cryptocurrency transactions using Ethereum smart contracts. The application enables users to connect their MetaMask wallet, enter the recipient's wallet address, specify the transaction amount, optionally include a message, and initiate transactions directly through the blockchain. The frontend is developed using React.js, Ethers.js is used to establish communication between the frontend and the Ethereum blockchain, Solidity is used to implement the smart contract logic, and Hardhat serves as the development and deployment framework.

The primary objective of this project is to demonstrate the practical implementation of decentralized payment systems using blockchain technology while providing an understanding of smart contracts, cryptocurrency transactions, wallet integration, and Web 3.0 application development.

Literature Survey

Blockchain technology and decentralized applications (DApps) have gained significant attention in recent years due to their ability to provide secure, transparent, and trustless digital transactions. The rapid growth of cryptocurrencies and Web 3.0 has encouraged researchers to develop decentralized payment systems that eliminate intermediaries while improving security, privacy, and transaction efficiency. Smart contracts deployed on blockchain networks have become a key innovation for automating financial transactions and ensuring data integrity. This literature survey reviews several research papers and existing blockchain technologies that provide the conceptual foundation for the proposed decentralized cryptocurrency transaction system..

Paper 1: Ethereum: A Secure Decentralised Generalised Transaction Ledger (2014)

Wood (2014) introduced Ethereum as a decentralized blockchain platform capable of executing programmable smart contracts. Unlike Bitcoin, which primarily supports cryptocurrency transactions, Ethereum enables developers to create decentralized applications using Solidity-based smart contracts. The platform provides transparency, immutability, and decentralized execution, making it suitable for financial applications and digital asset management.

Ethereum's programmable blockchain has become the foundation of thousands of decentralized applications across finance, healthcare, gaming, and supply chain management. However, high transaction fees (gas fees) and network congestion remain significant challenges during periods of heavy network activity.

The proposed system utilizes Ethereum because of its mature ecosystem, extensive developer support, and compatibility with modern Web 3.0 technologies while focusing on implementing secure peer-to-peer cryptocurrency transactions.

Paper 2: Smart Contracts for Decentralized Applications (2018)

Atzei, Bartoletti, and Cimoli (2018) studied the implementation and security aspects of Ethereum smart contracts. Their research highlighted how smart contracts automate transactions without requiring

centralized authorities, thereby reducing operational costs and improving transaction transparency. The paper also discussed common vulnerabilities such as re-entrancy attacks and emphasized the importance of secure smart contract development.

The research demonstrated that properly designed smart contracts can significantly improve the reliability and efficiency of decentralized financial systems. However, once deployed, smart contracts cannot be modified easily, making careful testing and verification essential before deployment.

The proposed project adopts secure smart contract practices and uses Hardhat for testing and deployment to ensure reliable execution of cryptocurrency transactions.

Paper 3: A Survey on Blockchain-Based Decentralized Applications (2022)

Xu et al. (2022) presented a comprehensive survey of decentralized applications developed using blockchain technology. The study analyzed the architecture of Web 3.0 applications, wallet integration, consensus mechanisms, and blockchain scalability. It concluded that decentralized applications provide greater transparency, enhanced security, and improved user ownership compared to traditional centralized systems.

The survey also identified challenges such as scalability limitations, user adoption, and blockchain transaction costs. Despite these limitations, decentralized applications continue to grow rapidly due to increasing trust in blockchain-based ecosystems.

The proposed system builds upon these concepts by integrating MetaMask wallet authentication, Ethereum smart contracts, and React.js to create a secure and user-friendly decentralized payment application.

Comparison Table

System / Research Paper	Main Focus	Advantages	Limitations	Proposed System Improvement
Wood (2014)	Ethereum Blockchain Platform	Secure smart contracts and decentralized execution	High gas fees and network congestion	Lightweight implementation for academic demonstration
Atzei et al. (2018)	Smart Contract Security	Automated and transparent transactions	Smart contracts require extensive testing before deployment	Uses Hardhat for smart contract testing and secure deployment
Xu et al. (2022)	Blockchain-based DApps	Decentralization, transparency, and user ownership	Scalability challenges and transaction costs	Simple peer-to-peer crypto transaction system using MetaMask and Ethereum
MetaMask & Ethers.js	Wallet integration and blockchain communication	Easy wallet connection and real-time blockchain interaction	Requires browser wallet installation	User-friendly interface with seamless wallet connectivity
Proposed System	Blockchain-based cryptocurrency transaction system	Secure, transparent, decentralized, and intermediary-free transactions	Limited to Ethereum network	Practical Web 3.0 payment application suitable for academic implementation

Problem Statement

Traditional digital payment systems rely on centralized financial institutions such as banks and payment gateways to process and verify transactions. Although these systems are widely used, they often involve several limitations, including high transaction fees, delays in payment processing, dependency on intermediaries, and limited transparency. In cross-border transactions, these issues become more significant due to currency conversion, multiple intermediaries, and longer settlement times.

Another major challenge is the lack of user control over financial assets. Since centralized authorities manage transaction records and account information, users must trust third parties to securely process and store their data. Additionally, centralized systems are vulnerable to security threats, single points of failure, and unauthorized access, which may compromise transaction integrity and user privacy.

With the increasing adoption of blockchain technology, there is a growing need for a decentralized payment system that enables users to transfer cryptocurrency securely without relying on intermediaries. Such a system should provide transparency, immutability, enhanced security, and direct peer-to-peer transactions while allowing users to maintain full control over their digital assets.

Therefore, the proposed project aims to develop a blockchain-based cryptocurrency transaction system that addresses these challenges by implementing Ethereum smart contracts, MetaMask wallet integration, and a decentralized application (DApp). The system provides secure and transparent cryptocurrency transactions, ensuring faster processing, reduced transaction costs, and reliable execution through blockchain technology.

Proposed Methodology

The proposed system follows a decentralized architecture for secure cryptocurrency transactions using blockchain technology. The application integrates a React.js frontend with the Ethereum blockchain through MetaMask and Ethers.js, while Solidity smart contracts execute and record transactions securely..

1. Frontend Application

The frontend provides an interactive interface through which users can access the decentralized application. It is responsible for:

- Connecting the user's MetaMask wallet
- Accepting the receiver's wallet address
- Accepting the cryptocurrency amount
- Allowing users to enter an optional transaction message
- Displaying transaction history
- Showing wallet connection status

2. MetaMask Wallet Integration

MetaMask acts as the authentication gateway between the user and the Ethereum blockchain. It is responsible for:

- Connecting the user's Ethereum wallet
- Verifying user identity
- Digitally signing transactions
- Requesting user approval before transaction execution
- Sending approved transactions to the Ethereum blockchain

3. Smart Contract

The smart contract is developed using Solidity and deployed on the Ethereum blockchain. It is responsible for:

- Validating transaction requests
- Executing cryptocurrency transfers
- Recording transaction details on the blockchain
- Maintaining transparency and immutability
- Eliminating the need for third-party intermediaries

4. Transaction Process

The overall workflow of the system is as follows:

- User opens the DecentraPay application
- User connects the MetaMask wallet
- Wallet authentication is completed
- User enters the receiver's wallet address
- User specifies the transaction amount
- User optionally enters a transaction message
- User confirms the transaction through MetaMask
- Smart contract validates and executes the transaction
- Ethereum blockchain records the transaction
- Updated transaction history is displayed to the user

5. Technologies Used

The technologies proposed for implementation are:

- React.js
- Solidity
- Ethereum Blockchain
- MetaMask
- Ethers.js
- Hardhat
- JavaScript
- HTML5
- CSS3

Conclusion

The proposed **DecentraPay: A Blockchain-Based Secure Cryptocurrency Transaction System** demonstrates the practical implementation of Web 3.0 technologies by enabling secure and decentralized cryptocurrency transactions. The system integrates React.js, MetaMask, Ethers.js, Solidity, and the Ethereum blockchain to provide a transparent and efficient platform for peer-to-peer digital payments.

By implementing smart contracts, wallet authentication, and blockchain-based transaction processing, the system offers the following advantages:

- Secure and transparent cryptocurrency transactions
- Elimination of third-party intermediaries
- Enhanced user control over digital assets
- Immutable and tamper-proof transaction records
- Faster transaction processing with improved reliability

The project also provides practical knowledge of blockchain development, decentralized applications, smart contract deployment, and wallet integration. It serves as an academic implementation of Web 3.0 concepts while demonstrating the potential of blockchain technology in modern financial systems.

Future enhancements may include:

- Multi-chain blockchain support
- Integration with decentralized finance (DeFi) services
- QR code-based cryptocurrency payments
- Mobile application development
- Layer-2 blockchain solutions for reduced transaction costs and improved scalability

The proposed system demonstrates how blockchain technology can be used to build secure, transparent, and decentralized payment applications suitable for real-world digital transactions.

References

1. Wood, G. (2014). *Ethereum: A Secure Decentralised Generalised Transaction Ledger*. Ethereum Foundation. https://ethereum.github.io/yellowpaper/paper.pdf
2. Buterin, V. (2014). *A Next-Generation Smart Contract and Decentralized Application Platform (Ethereum Whitepaper).* https://ethereum.org/en/whitepaper/
3. Solidity Documentation. Ethereum Foundation. https://docs.soliditylang.org/
4. Hardhat Documentation. Nomic Foundation. https://hardhat.org/docs
5. Ethers.js Documentation. https://docs.ethers.org/
6. MetaMask Developer Documentation. Consensys. https://docs.metamask.io/
7. Ethereum Official Documentation. Ethereum Foundation. https://ethereum.org/en/developers/docs/
8. Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.* https://bitcoin.org/bitcoin.pdf
9. Xu, X., Weber, I., & Staples, M. (2019). *Architecture for Blockchain Applications*. Springer Nature. https://link.springer.com/book/10.1007/978-3-030-03035-3
10. React Documentation. Meta Open Source. https://react.dev/